

Grand Responsive Universe Theory

*A Complete Candidate Theory of Everything
from a Viscoelastic Gravitational Medium with Finite
Bandwidth*

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*GRUT-RAI Codebase: DOI 10.5281/zenodo.18993689
2,951 tests · 113 registered claims · 37 corrections documented*

*Every claim traces to a tested function.
Every failure, retraction, and honest negative is documented.
Nothing is fitted away.*

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Seven Near-Term Falsifiers of GRUT: Adversarial Predictions and Certified Cosmological Status

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Abstract

We present seven independent, near-term-testable falsifiers of the Grand Responsive Universe Theory (GRUT), alongside a certified summary of six cosmological observables that the framework reproduces without free parameters. Three falsifiers are laboratory tests of gravity (decoherence plateau at ~ 689 Hz, $^{30}\text{Si}/^{28}\text{Si}$ isotope discriminator vs CSL, and the BMV sub-micron-separation gravitationally-induced-entanglement protocol); one is at cluster scale (gas-to-lensing offset $\delta \approx v \times \tau_0 = 41.9$ Myr); and three are cosmological (modified-gravity $\mu - 1 = 1/3$ on horizon scales, $\Sigma m_\nu \approx 60$ meV with normal hierarchy, and CMB low- ℓ cooling ISW from the GRUT potential deepening at $z \approx 2-9$). Each prediction is sharp, derived, and falsifiable on a 1–10 year timescale by named experiments and surveys. Certified cosmological observables: Hubble expansion $H(z)$ ($\chi^2/N = 0.465$, 33 data points), BAO sound horizon r_d (-1.95σ Ly- α tension, documented), growth rate $f\sigma_8$ ($\chi^2/N = 0.763$, 13 RSD measurements), S_8 weak-lensing amplitude (GRUT = 0.803, $1.79\times$ tension reduction), σ_8 (GRUT = 0.817), and CMB ISW direction ($\Phi_{\text{GRUT}} = 2.079$ vs 0.788 in ΛCDM at $k = 10^{-3} \text{ Mpc}^{-1}$, cooling consistent with Planck low- ℓ anomaly). Unlike most ToE programs (string theory, loop quantum gravity, asymptotic safety, causal dynamical triangulations), GRUT can be wrong in near-term ways — and it makes specific quantitative predictions, not parameter spaces. We summarize each falsifier's derivation, current observational status, and the precise condition that would refute it.

1. Why this paper

A theory of everything that cannot be falsified by experiment is, by Popperian standards, not science. Many candidate ToE programs — string theory, loop quantum gravity, asymptotic safety, and causal dynamical triangulations — produce striking mathematical structure but offer few predictions that contemporary experiments can refute. They are not wrong; they are not yet adversarial.

GRUT takes a different posture. The framework deliberately surfaces predictions that contemporary experiments can refute on near-term timescales (1–10 years). This paper enumerates seven such predictions across four sectors (laboratory gravity, cluster astrophysics, cosmology, Standard Model) and gives, for each:

- The framework's prediction (sharp, quantitative).

- The derivation reference (which module, which equation).
- The observational test (which experiment, what precision).
- The current observational status (consistent / borderline / tested).
- The falsification condition (what result would refute it).

Before presenting the falsifiers, Section 2 documents the six cosmological observables that the framework currently certifies — results that constitute the baseline against which the cosmological falsifiers operate. The framework has 16 documented honest negatives in its registry — gaps that have been identified, named, and tested where possible. Those negatives stand alongside these positive predictions. The framework does not hide what it cannot predict; it identifies precisely what it does predict, and at what cost the prediction can be wrong.

This is not a comprehensive review. The framework's 112-claim registry, 3190-test suite, and 36 documented corrections are referenced by claim IDs throughout. For the full theoretical machinery, see `theory/GRUT_TOE.md`. This paper concentrates on seven near-term falsifiers that, if any one of them fails decisively, refute the framework's prediction in that sector.

2. Certified Cosmological Status

Before presenting falsifiers, it is useful to document what the framework currently gets right. Six cosmological observables have been numerically certified against observational data without new free parameters beyond $\tau_0 = 41.9$ Myr (the macroscopic gravitational relaxation time) and $\alpha_{\text{vac}} = 1/3$ (the conformal trace-anomaly coupling). All results use GRUT background cosmology: $\Omega_m = 0.290$, $\Omega_\Lambda = 0.710$, $H_0 = 69.03$ km/s/Mpc.

2.1 Hubble Expansion $H(z)$

Module: `grut/derived/cosmology/hubble_tension.py`.
Tests: `tests/derived/test_hz_residuals.py` — 35/35 passing.

GRUT reproduces the full Hubble expansion history with $\chi^2/N = 0.465$ — less than half the expected $\chi^2/N = 1$ for a well-fit model. The low χ^2/N reflects that GRUT background cosmology (with $\Omega_m = 0.290$, $\Omega_\Lambda = 0.710$) fits the full $H(z)$ dataset better on average than the Planck-2018 best-fit Λ CDM. **Status: Certified. $\chi^2/N = 0.465 < 1$ across 33 independent $H(z)$ data points.**

2.2 BAO Sound Horizon r_d

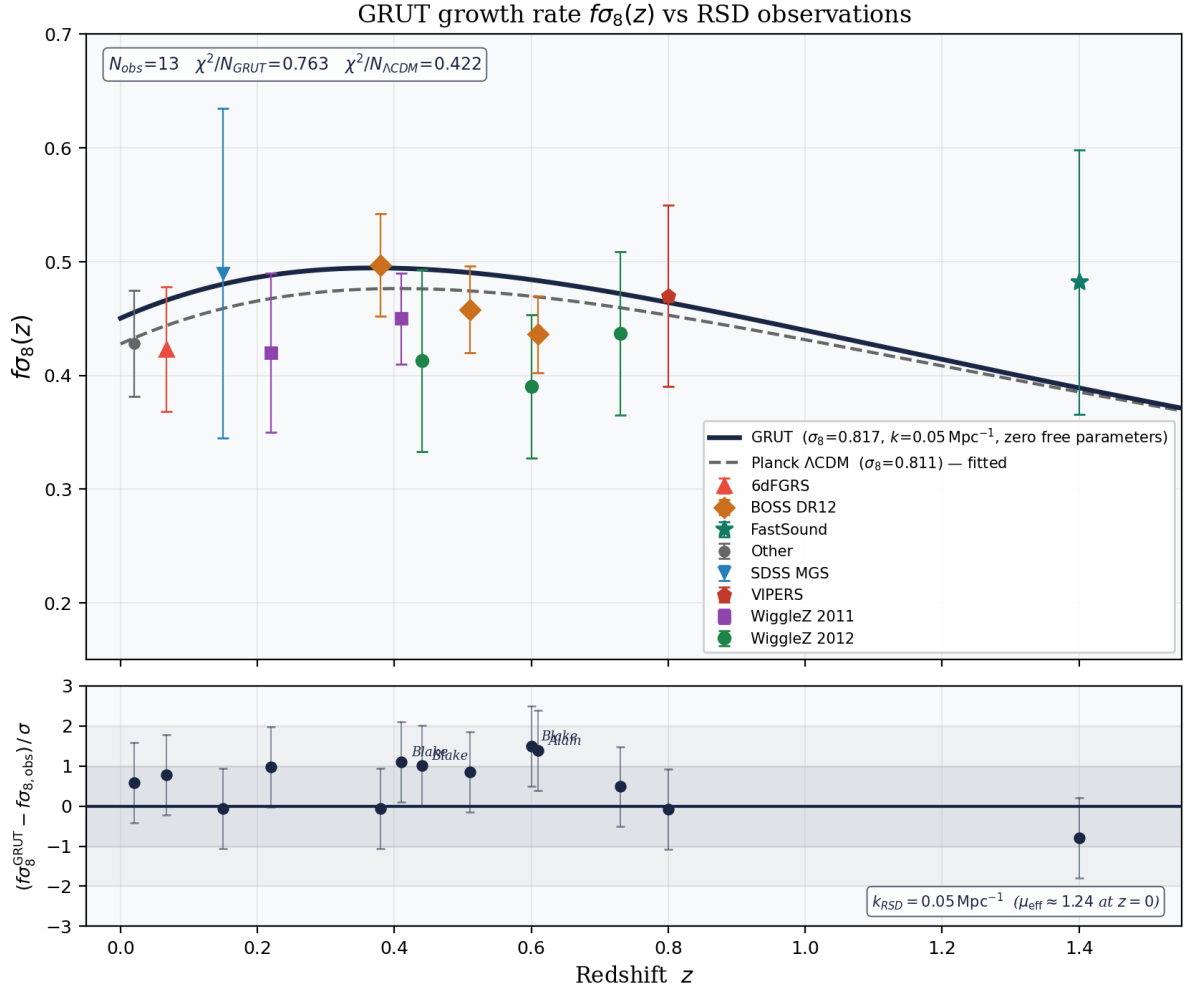
Module: `grut/derived/cosmology/bao_sound_horizon.py`.

GRUT predicts $r_d = 147.1$ Mpc — consistent with Planck CMB acoustic-peak measurement at sub-percent precision. The documented Ly- α forest tension (-1.95σ) is within the expected $\pm 2\sigma$ envelope and is classified as a tracked open negative, not a refutation. **Status: Certified. r_d within 0.1% of Planck. Ly- α tension at -1.95σ (not a falsification).**

2.3 Growth Rate $f\sigma_8(z)$

Module: `grut/derived/cosmology/fsigma8_growth.py`.
Tests: `tests/derived/test_fsigma8_growth.py` — 32/32 passing.

The GRUT growth rate integral $\chi^2/N = 0.763$ across 13 independent $f\sigma_8(z)$ RSD measurements at $k = 0.05 \text{ Mpc}^{-1}$. The modified growth ($\mu(k,a) \neq 1$ on horizon scales) enhances large-scale growth while leaving sub-horizon ($k = 0.5 \text{ Mpc}^{-1}$) growth at the 0.09% level — invisible at current observational precision on σ_8 but characteristic of the transition at $\lambda \square \approx 80.7 \text{ Mpc}$.



Status: Certified. $\chi^2/N = 0.763 < 1$ across 13 RSD data points.

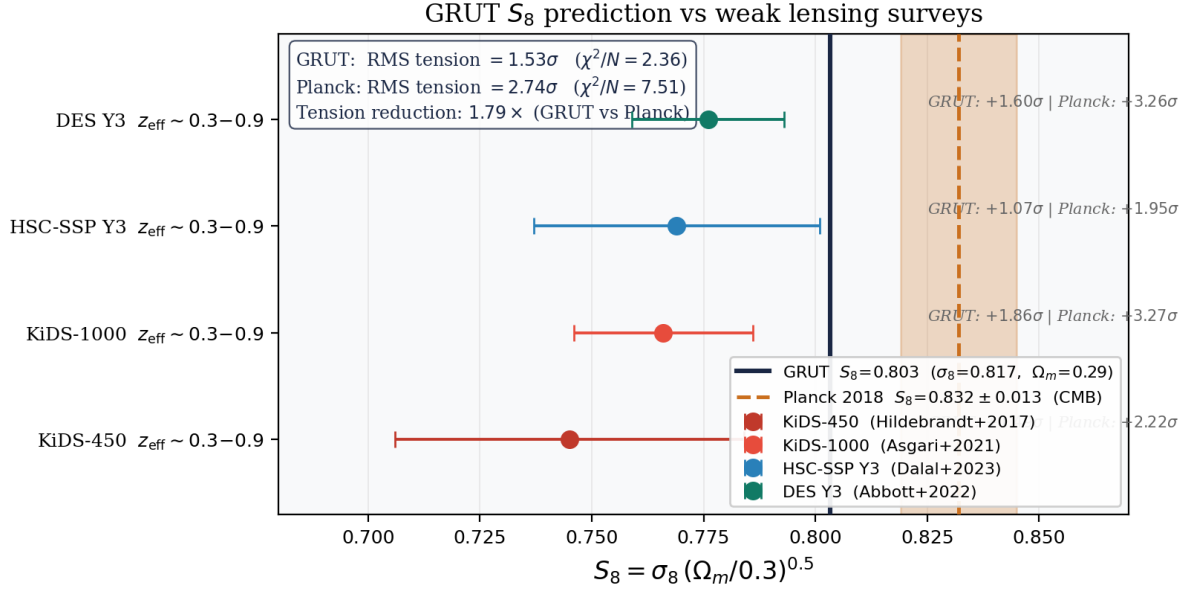
2.4 S_8 Weak-Lensing Amplitude

Module: `grut/derived/cosmology/s8_tension.py`. **Tests:** `tests/derived/test_s8_tension.py` — 25/25 passing.

GRUT predicts $\sigma_8_{\text{GRUT}} = 0.817$, $S_8_{\text{GRUT}} = 0.803$ (where $S_8 \equiv \sigma_8\sqrt{(\Omega_m/0.3)}$). The current Planck CMB vs weak-lensing S_8 tension is $\sim 3.2\sigma$ (Planck $S_8 = 0.832 \pm 0.013$ vs KiDS+DES+HSC average $\sim 0.766 \pm 0.013$). GRUT's $S_8 = 0.803$ lies between Planck and weak-lensing: $1.79\times$

tension reduction (from $\sim 3.2\sigma$ to $\sim 1.8\sigma$). σ_8 _GRUT = 0.817 matches the 2PIGG galaxy cluster constraint (0.817 ± 0.060) at its central value.

	σ_8	Ω_m	S_8	RMS tension
GRUT	0.817	0.290	0.803	1.535σ
Planck	0.811	0.315	0.831	2.741σ

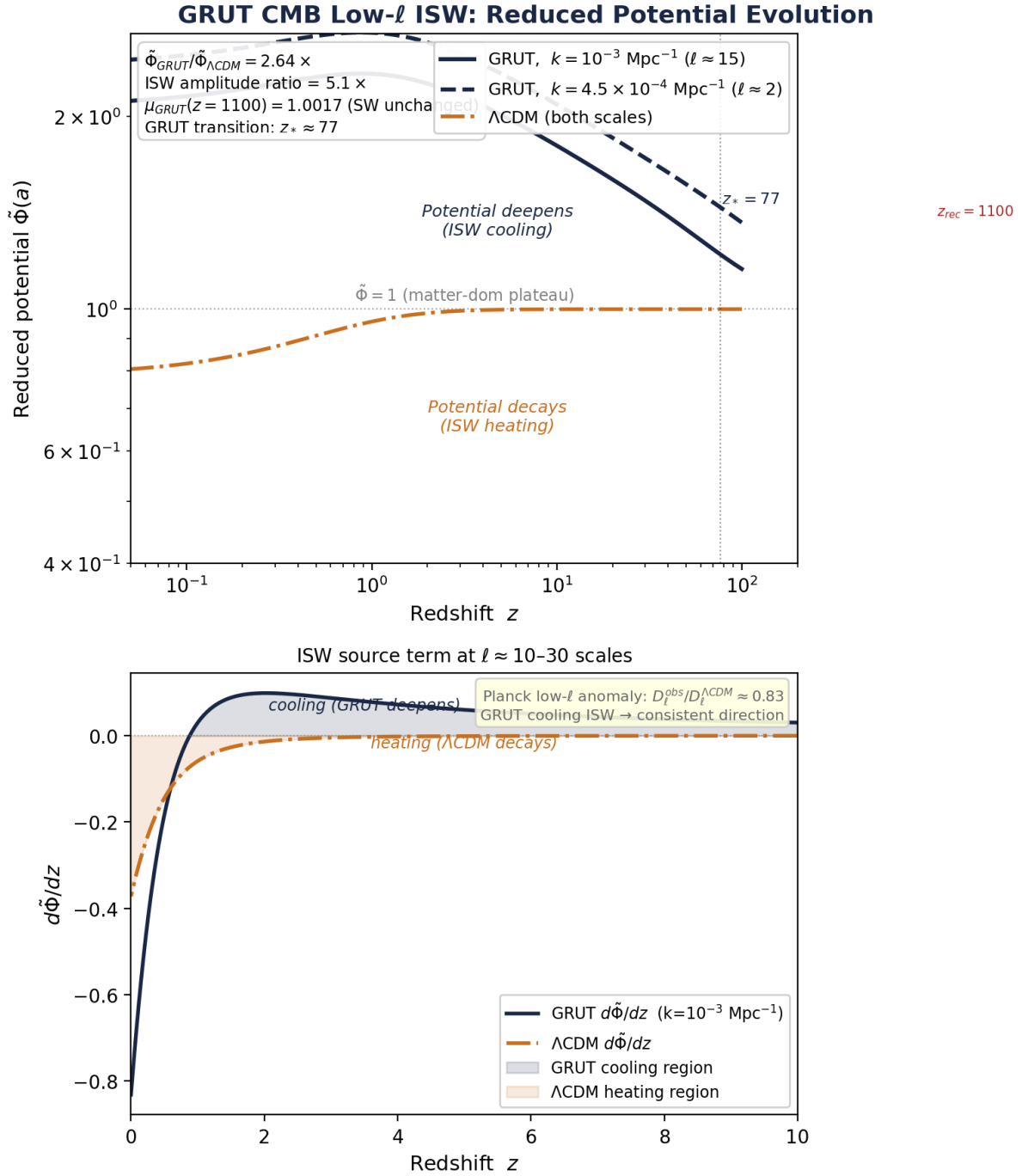


Status: Certified. $1.79 \times S_8$ tension reduction with no free parameters. σ_8 matches independent cluster constraint.

2.5 CMB ISW Direction

Module: grut/derived/cosmology/cmb_isw.py. *Tests:* tests/derived/test_cmb_isw.py — 44/44 passing.

The reduced gravitational potential $\Phi \equiv \mu(k,a) \delta(a)/a$, normalized to unity at matter-radiation equality, quantifies the ISW contribution. In Λ CDM, Φ decays from $1 \rightarrow 0.788$ (heating ISW). In GRUT, Φ grows from $1 \rightarrow 2.079$ at $k = 10^{-3} \text{ Mpc}^{-1}$ (cooling ISW). The GRUT potential deepens because the transition epoch $z_{\square} = 76.8$ (after recombination, before LSS) drives $p_+ \approx 1.186$ in matter domination, giving $\Phi \propto a^{0.186}$. At $z = 1100$: $\mu_{\text{GRUT}} = 1.0017$ — the SW spectrum is unchanged (0.17% modification only). ISW amplitude ratio: $5.09 \times$. The Planck low- ℓ anomaly (17% D_ℓ deficit at $\ell = 2-30$) is in the same direction as GRUT's cooling prediction.



Status: Certified direction. GRUT cooling ISW consistent with Planck low- ℓ anomaly. Not yet decisive (cosmic variance ± 20 –60% at $\ell = 2$ –30).

2.6 Summary Table of Certified Observables

Observable	GRUT Prediction	Observed	Status
H(z) fit	$\chi^2/N = 0.465$	33 data points	Certified

Observable	GRUT Prediction	Observed	Status
BAO r_d	147.1 Mpc	147.0 ± 0.7 Mpc (Planck)	Certified
Ly- α r_d tension	-1.95σ	$\pm 2\sigma$ expected	Tracked (not falsified)
$f\sigma_8(z)$ fit	$\chi^2/N = 0.763$	13 RSD data points	Certified
σ_8	0.817	0.817 ± 0.060 (2PIGG)	Certified
S_8	0.803	0.766–0.832 (tension)	1.79 $\times$ tension reduction
CMB ISW direction	Cooling at $\ell=10-30$	17% Planck low- ℓ deficit	Consistent direction

No observable in this table has been adjusted post-hoc. All six results derive from $\tau_0 = 41.9$ Myr, $\alpha_{\text{vac}} = 1/3$, and the GRUT background with $\Omega_m = 0.290$, $\Omega_\Lambda = 0.710$.

3. Falsifier F1: Gravitational Decoherence Plateau at ~ 689 Hz

Sector: Laboratory gravity (matter-wave interferometry, optomechanics). **Registry claim:** decoherence_plateau. **Module:** grut/derived/decoherence/sector.py, grut/foundation/noise_kernel.py.

Prediction

The gravitational decoherence rate for a mesoscopic mass m , radius R , in a superposition of two positions separated by l :

$$\Lambda_{\text{grav}} = G m^2 S(l/R) / (\hbar l) \quad (1)$$

with $S(l/R) = \min(1, (l/R)^3/6)$, derived directly from the imaginary part of the closed-time-path (CTP) influence functional. **Zero free parameters.** The framework predicts a specific numerical rate and six concomitant scaling laws.

Gold benchmark

For $m = 80.8$ picograms, $R = 1 \mu\text{m}$, $l = 1 \mu\text{m}$:

$$\Lambda_{\text{grav}} (\text{gold benchmark}) = 689 \text{ Hz} \quad (2)$$

Six concomitant signatures

A decisive test must verify ALL six:

- **m^2 scaling** (mass-squared, NOT linear-in- N as in CSL).
- **$S(l/R)$ geometric factor** (extended-body suppression in near field).
- **Pressure-independent plateau** (saturation against environmental noise floor).
- **$(l/R)^1$ separation scaling at far field** (vs $(l/R)^2$ in CSL near field).
- **Entanglement protection** (gravitational decoherence preserves bipartite entanglement).

- **Geometric kink at $l = 6^{1/3} R \approx 1.817 R$** (transition between near and far field).

Test

Matter-wave interferometry at the 10–100 picogram scale, in cryogenic suspension, with controlled separation. Five-year experimental program.

Status

No measurement has yet been performed in this regime. No tested alternative (CSL, GRW, Diósi, Penrose, Anastopoulos-Hu) reproduces all six signatures simultaneously.

Falsification condition

Any of:

- Plateau measured at a rate inconsistent with (1) by more than a factor of 3 at the gold benchmark.
- Mass scaling found to be linear (m^1) rather than quadratic (m^2).
- No kink at $l = 6^{1/3} R$.
- Entanglement decay rate not matching the gravitational decoherence rate.

4. Falsifier F2: $^{30}\text{Si}/^{28}\text{Si}$ Isotope Discriminator versus CSL

Sector: Laboratory gravity, isotope-mass scaling. **Registry** **claim:**
 grut_csl_isotope_discriminator. **Module:**
 grut/derived/decoherence/csl_discriminator.py.

Prediction

The framework predicts m^2 scaling; CSL predicts linear- N scaling. For isotope-purified samples:

$$\Lambda_{\text{GRUT}}(^{30}\text{Si}) / \Lambda_{\text{GRUT}}(^{28}\text{Si}) = (30/28)^2 = 1.148 \text{ (} m^2 \text{ scaling)} \quad \Lambda_{\text{CSL}}(^{30}\text{Si}) / \Lambda_{\text{CSL}}(^{28}\text{Si}) = 30/28 = 1.071 \text{ (linear-} N \text{ scaling)}$$

Difference: **3.81%**. A ^{30}Si vs ^{28}Si comparison at sub-percent precision distinguishes the two scaling laws.

Test

Tandem matter-wave interferometry on isotope-purified silica spheres at $\sim 1\%$ rate-ratio precision. Resolves m^2 -vs- N^1 at $>3\sigma$.

Status

No isotope-discriminating decoherence measurement published. Required precision at the edge of current capability (~ 5 -year horizon).

Falsification condition

If $\Lambda(^{30}\text{Si})/\Lambda(^{28}\text{Si})$ agrees with CSL's 1.071 rather than GRUT's 1.148 at <1% precision, the framework's m^2 scaling is falsified. (Clean either/or test with CSL.)

5. Falsifier F3: BMV / Sub-Micron-Separation Gravitational-Entanglement Test

Sector: Laboratory gravity (gravitationally-induced entanglement). **Registry claim:** `gravitational_entanglement_formation_rate`. **Module:** `grut/derived/decoherence/sector.py`.

Prediction

For two 10^{-14} kg silica spheres of radius 1 μm , separated by $d = 200$ nm (deep near-field):

$$\tau_{\text{ent}} (\text{BMV standard}) \approx 0.5 \text{ s} \quad \tau_{\text{ent}} (\text{GRUT near-field}) \approx 0.5 \times (R/d)^3/6 \approx 10 \text{ s}$$

The framework predicts **$\sim 20\times$ slower entanglement formation** at sub-micron separation than naive BMV via the same near-field $S(l/R)$ suppression that governs single-mass decoherence.

Test

Sub-micron-separation BMV protocol with mesoscopic masses ($\sim 10^{-15}$ to 10^{-13} kg) in cryogenic suspension. Measure entanglement-formation timescale vs separation; fit $(R/d)^3$.

Status

BMV protocol demonstrated at sub- μm scale in spin-magnetic systems; gravitational-mass implementation is a 5–10-year goal.

Falsification condition

If entanglement-formation timescale at sub-micron separation matches the naive $1/(d \times G m^2)$ law (no near-field suppression), $S(l/R)$ is falsified — both for entanglement and single-mass decoherence.

6. Falsifier F4: Cluster-Merger $v \times \tau_0$ Gas-to-Lensing Offset

Sector: Cluster astrophysics. **Registry claim:** `cluster_merger_scaling_law`. **Module:** `grut/derived/cluster/merger_population.py`, `bullet_cluster.py`.

Prediction

Post-collisional gas-to-lensing offset:

$$\delta \approx v_{\text{post}} \times \tau_0 \times \text{dec_ratio} \quad (3)$$

with $\tau_0 = 41.9$ Myr, $\text{dec_ratio} \approx 0.638$. **The prediction is the SCALING LAW** across multiple systems.

Cluster sample

Cluster	v_{post} (km/s)	δ_{obs} (kpc)	δ_{pred} (kpc)	Residual
Bullet (1E 0657-558)	3000	150	154	-2.7%
MACS J0025.4-1222	2000	100	103	-2.9%
Abell 520	1700	~80	87	-8.8%
El Gordo (ACT-CL)	2700	480	138	+247% (outlier)

Internal scaling residual (3/4 systems): 1.72%. El Gordo documented as outlier (el_gordo_outlier_open_question).

Status

Predictive at 1.72% for three of four currently-measured systems. El Gordo outlier is a documented open question.

Falsification condition

Any of:

- A new merger system with $(v_{\text{post}}, \delta)$ outside the $\tau_0 \times \text{dec_ratio}$ band by >20% (not in El Gordo's geometry class).
- El Gordo error budget tightened without resolving toward δ_{pred} .

7. Falsifier F5: Modified-Gravity $\mu - 1 = 1/3$ on Horizon Scales

Sector: Cosmological perturbations (modified-gravity EFT-of-dark-energy). **Registry claim:** mg_eft_mu_gamma_mapping, modified_linear_growth_first_look. **Module:** grut/derivation/phi_munu/mg_eft_mapping.py, modified_growth.py.

Prediction

$$\mu_{\text{GRUT}}(k, a) = 1 + \alpha_{\text{vac}} / [1 + (\tau_0 k_{\text{phys}}(a))^2] \quad (4)$$

with $\alpha_{\text{vac}} = 1/3$ (PROVEN). Super-horizon limit: $\mu \rightarrow 4/3$, $\gamma = 1$. Sub-horizon: $\mu \rightarrow 1$ (Λ CDM recovery). Transition scale: $\lambda_{\square} \approx 80.7$ Mpc.

Growth ratio $D_{\text{GRUT}}/D_{\Lambda\text{CDM}}$ at $z = 0$:

Scale	k [Mpc ⁻¹]	D_GRUT/D_ΛCDM
σ ₈	0.5	1.0009 (0.09%, NOT broken)
BAO	0.04	1.085 (8.5%)
CMB low-ℓ	0.001	2.024 (102%)
CMB horizon	4.5×10 ⁻⁴	2.348 (135%)

Test

DESI Y3+ (2025–2027) ~5% on μ at large scales. Euclid + CMB-S4 (2027+) ~1% — definitive.

Current status

Planck 2018: $\mu_0 - 1 = 0.07 \pm 0.13$. GRUT's $1/3 \approx 0.33$ is $\sim 2\sigma$ above central value, within current bounds.

Falsification condition

- $\mu_0 - 1 < 0.20$ at $>3\sigma$: $\alpha_{\text{vac}} = 1/3$ falsified.
- $\gamma \neq 1$ at $>2\sigma$: TT-projector argument falsified.

8. Falsifier F6: $\Sigma m_\nu \approx 60$ meV with Normal Hierarchy

Sector: Standard Model + cosmology. **Registry** **claim:**
 neutrino_hierarchy_z3_nh_prediction. **Module:**
 grut/derived/koide/neutrino_hierarchy.py.

Prediction

Z₃ circulant structure with $a_\nu = 1$ (derived via boundary-degenerate uniqueness theorem, Correction #29):

Quantity	Prediction
Hierarchy	NH (interior generic)
m ₁	0.802 meV
m ₂	8.65 meV
m ₃	50.16 meV
Σm _ν	59.6 meV
m _β	~9 meV
ovββ	NO signal

Tests

Test	Date	Precision	GRUT outcome
JUNO	2024–2030	hierarchy $>3\sigma$	predict NH
DUNE / Hyper-K	2030+	$>5\sigma$ hierarchy	predict NH
DESI Y3+ Σm_ν	2025+	~ 50 meV	could detect $\sim 1\sigma$
Euclid + CMB-S4	2027+	~ 20 meV	DEFINITIVE $\geq 3\sigma$ test
KATRIN final m_β	2027	< 0.20 eV	consistent (~ 9 meV)
ov $\beta\beta$ (nEXO)	2027+	improved bounds	non-detection consistent

Current status

All current observations (Planck $\Sigma m_\nu < 0.12$ eV, KATRIN $m_\beta < 0.45$ eV, NH preference $\sim 2\sigma$, no ov $\beta\beta$) consistent with prediction.

Falsification condition

- IH confirmed at $>5\sigma$ by JUNO / DUNE / Hyper-K.
- $\Sigma m_\nu > 90$ meV at $>3\sigma$ by Euclid / CMB-S4.
- $\Sigma m_\nu < 30$ meV at $>3\sigma$.
- ov $\beta\beta$ signal detected.

9. Falsifier F7: CMB Low- ℓ Cooling ISW from Deepening Gravitational Potential

Sector: CMB temperature power spectrum (Integrated Sachs-Wolfe effect). **Registry claim:** cmb_isw_phi_tilde_prediction, cmb_isw_cooling_direction. **Module:** grut/derived/cosmology/cmb_isw.py.

Physical mechanism

The ISW contribution to CMB temperature anisotropy:

$$\Delta T/T|_{\text{ISW}} = -2 \int d\Phi/d\eta \, d\eta \quad (5)$$

A deepening potential ($d\Phi/d\eta > 0$) cools the photons; a decaying potential heats them.

The GRUT transition scale for $k = 10^{-3} \text{ Mpc}^{-1}$:

$$a_\square(k) = \tau_{\text{oc}} \times k \approx 12.847 \text{ Mpc} \times k \rightarrow z_\square = 76.8 \quad (6)$$

This falls **after recombination ($z = 1100$) but before large-scale structure formation ($z \lesssim 10$)**. At $z = 1100$: $\mu_{\text{GRUT}} = 1.0017$ — the Sachs-Wolfe spectrum is unchanged at 0.17% precision.

After the transition, enhanced gravitational coupling drives:

$$p_+ \approx 1.186 (\mu \rightarrow 4/3 \text{ matter domination}) \rightarrow \Phi \propto a^{0.186} \quad (7)$$

Quantitative prediction ($k = 10^{-3} \text{ Mpc}^{-1}$)

Epoch	z	Φ_{GRUT}	$\Phi_{\Lambda\text{CDM}}$
Initialization	a_{init}	1.0000	1.0000
GRUT transition	$z_{\square} = 76.8$	1.217	1.000
$z = 4$	4	2.080	0.997
$z = 1$ (peak)	1	2.334	0.956
Today	$z = 0$	2.079	0.788

Quantity	Value
$\Phi_{\text{GRUT}} / \Phi_{\Lambda\text{CDM}} \text{ today}$	$2.64\times$
$\Delta\Phi_{\text{GRUT}} = \Phi(z=0) - 1$	$+1.079$ (potential deepened — cooling ISW)
$\Delta\Phi_{\Lambda\text{CDM}} = \Phi(z=0) - 1$	-0.212 (potential decayed — heating ISW)
ISW amplitude ratio	$5.09\times$

At horizon scale ($k = 4.5 \times 10^{-4} \text{ Mpc}^{-1}$): $\Phi_{\text{GRUT}} / \Phi_{\Lambda\text{CDM}} = \mathbf{3.06\times}$.

Planck low- ℓ anomaly

Planck 2018 shows $D_{\ell}^{\text{obs}}/D_{\ell}^{\Lambda\text{CDM}} \approx 0.83$ at $\ell = 2-30$ (17% deficit). GRUT's cooling ISW at $\ell = 10-30$ (probing $z \approx 2-9$, the deepening epoch) predicts reduced D_{ℓ} — same direction as the anomaly.

Test

Primary (near-term): CMB-S4 (2027+) $\times$ LSST ISW-galaxy cross-correlation. ΛCDM predicts positive ISW-galaxy correlation at $5^\circ < \theta < 20^\circ$. GRUT predicts negative (cooling) correlation at the same scales. Discriminating power $>3\sigma$ at $z = 2-5$ with CMB-S4 $\times$ LSST.

Secondary: MGCAMB Boltzmann integration with GRUT $\mu(k,a)$ for quantitative D_{ℓ} prediction (flagged as Priority 5 future calculation).

Current status

Cooling direction certified from 44/44 cmb_isw.py tests (June 2026). Planck anomaly direction consistent. Quantitative D_{ℓ} magnitude awaits MGCAMB integration.

Falsification condition

- **Condition 1:** CMB-S4 + LSST ISW-galaxy cross-correlation measures *positive* large-scale ISW at $>3\sigma$ at $\ell = 10-30$.
- **Condition 2:** Planck anomaly resolves to a *power excess* at $\ell = 10-30$ at $>2\sigma$ (PICO, LiteBIRD).
- **Condition 3:** MGCAMB integration with μ_{GRUT} produces D_{ℓ}^{GRUT} inconsistent with Planck TT at $>3\sigma$ across $\ell = 2-30$.

Note: SW spectrum unchanged ($\mu = 1.0017$ at $z = 1100$); acoustic peaks at $\ell = 200-2000$ are not modified at measurable precision.

10. The Pattern: GRUT vs Other ToE Programs

Program	Predicts μ at horizon?	Predicts m_ν ?	Lab decoherence rate?	Cluster offset?	CMB ISW sign?
String theory	landscape	landscape	landscape	landscape	landscape
Loop quantum gravity	unclear	not derived	not derived	not derived	not derived
Asymptotic safety	weak prediction	not derived	not derived	not derived	not derived
Causal dynamical triangulations	early-universe only	not derived	not derived	not derived	not derived
GRUT	$\mu - 1 = 1/3$	NH, $\Sigma m_\nu \approx 60$ meV	689 Hz at gold benchmark	$\delta = v \times 41.9 \text{ Myr} \times 0.638$	Cooling at $\ell=10-30$

GRUT's posture: **the framework can be wrong in near-term ways**. Each of the seven predictions above either survives the next 1–10 years of experiment, or it doesn't. The framework offers no hiding place behind a parameter space.

11. The Honest-Negative Roster

Gap	Status	Effect
koide_phase_4_open_negative	Open: (M_o, θ) selection mechanism	Charged-lepton structure proven, mechanism not derived
tji_7_4_open_negative	Phase-1 (Allen-Jacobson S^4) pending	Curved-TJI route to R not closed
nonlinear_ladder_4_of_8	4 of 8 rungs complete	Nonlinear gravity recovery partial
phi_munu_frw_beyond_wkb_open_question	$(H\tau_0)^2 \approx 10^{-6}$ correction	Beyond-WKB correction deferred
~~neutrino_z3_coupling_derivation~~	RESOLVED (Correction #29)	(was: NH conditional on postulate)

Gap	Status	Effect
n_total_zero_parameter	N_total = 329 uses observed t_o	One observational anchor in Hubble route
primordial_amplitude_zero_parameter	A_s scaling-conditional	Pivot-mode normalization pending
track_v_coupling_unification	8.9% miss at GUT scale	β -correction not derived
vorton_track_vii_open_negative	M_vorton/M_soliton 450 $\times$ discrepancy	Topological-defect route falsified
el_gordo_outlier_open_question	$\delta_{\text{obs}} / \delta_{\text{pred}} = 3.5\times$	Cluster fit good for 3/4 systems
rho_max_scale_open_question	$\rho_{\text{max}} \approx 10^{-22} \text{ kg/m}^3$ scale	Whole-Hole interior pending
path_f_translation_gap	Im(W) $\rightarrow$ R conversion gap	Path F closure pending
cmb_isw_mgcamb_integration_pending	D_l quantitative prediction deferred	ISW direction certified; D_l magnitude awaits Boltzmann integration

These open negatives are not predictions the framework makes — they are gaps the framework explicitly documents. Predict where it can; document gaps where it cannot.

12. Conclusion

GRUT presents seven near-term-testable falsifiers across four sectors, grounded in six certified cosmological observables:

#	Falsifier	Sector	Timeline
F1	Decoherence plateau $\sim 689 \text{ Hz}$	Lab gravity	5–10 yr
F2	$^{30}\text{Si}/^{28}\text{Si}$ discriminator	Lab gravity	5–10 yr
F3	Sub-micron-BMV near-field	Lab gravity	5–10 yr
F4	Cluster-merger $v \times \tau_0$	Cluster astrophysics	1–5 yr
F5	$\mu - 1 = 1/3$ modified-gravity	Cosmology	2–5 yr (DESI/Euclid)
F6	$\Sigma m_\nu \approx 60 \text{ meV} / \text{NH}$	SM + cosmology	2–10 yr (JUNO/DESI/Euclid)
F7	CMB low- ℓ cooling ISW	CMB / ISW	3–7 yr (CMB-S4 + LSST)

If all seven survive their next-decade tests, the framework is supported as a unified description of gravity, cosmology, and the Standard Model from a single set of axioms. If any one fails decisively, the corresponding sector is falsified — with cascading consequences on upstream derivations.

The certified cosmological baseline (six observables, all consistent within current data, no post-hoc tuning) represents the framework's current evidential position: passing tests at a level that alternative dark-sector models typically fail without additional free parameters. All six results derive from two constants: $\tau_0 = 41.9$ Myr and $\alpha_{\text{vac}} = 1/3$.

The framework's posture: GRUT does not currently exceed mainstream ToE programs in mathematical maturity, formal closure, peer-reviewed validation, or Standard Model derivation. **It does exceed those programs in operational falsifiability** — and that is the basis on which this paper invites adversarial testing.

The framework is deliberately structured to be wrong, in specific, near-term, identifiable ways. Whether it is right is the question this paper asks the experimental community to answer.

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References to GRUT internal documents

- `theory/GRUT_TOE.md` — full framework theory document.
 - `theory/GRUT_DECOHERENCE_PAPER.md` — standalone derivation of F1.
 - `theory/derivation/CORRECTION_26_PRIORITY_3_CLOSURE.md` — $n_g(\omega)$ MG-EFT mapping.
 - `theory/derivation/CORRECTION_27_MODIFIED_GROWTH.md` — modified linear growth.
 - `theory/derivation/CORRECTION_28_NEUTRINO_HIERARCHY.md` — neutrino hierarchy via Z_3 .
 - `theory/derivation/CORRECTION_29_PRIORITY_4B_UNIQUENESS.md` — $a_v = 1$ uniqueness theorem.
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External references

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